GEOLOGY IN HISTORY

4.2 How Geology Shapes the Battlefield

MAIN IDEA Geologic forces shape the landscape, impacting the outcome of many battles.

By Andrés Ruzo, **National Geographic Explorer**

TAKE THE HIGH GROUND

Taking advantage of a battlefield can mean the difference between victory and defeat, and there is a geologic story behind every great battle. An area's **geology**—the local geologic features such as rocks, landforms, and the processes that created and shaped them—can define a battlefield. Over millions, even billions of years, colliding tectonic plates and volcanic activity give rise to great landforms like mountain ranges. Erosion from weathering, moving glaciers, and other destructive forces break down these landforms. The eroded material is transported and re-deposited, customizing each region with its distinctive **terrain**, the physical features of the land.

Throughout history, successful commanders were the ones who understood how to use the local terrain to their advantage. The names of American Revolution battle sites like Brooklyn Heights (New York), Bunker Hill (Massachusetts), and Kings Mountain (South Carolina) are clues to the importance of battlefield geology. Hills, cliffs, bluffs, and other elevated landforms played a key role in the war. Elevation gave soldiers a strategic military advantage. Think about running uphill with a heavy rifle straight into enemy gunfire. Now consider the benefits of firing down on your enemies from a well-fortified position as they advance up difficult terrain. It's no wonder military commanders try to "take the high ground."

BUNKER HILL

Study the map of Bunker Hill. This landscape tells a geologic story. Over 20,000 years ago, a massive melting glacier slowly began to reveal the landscape that would one day become Boston. Geologically, Bunker Hill and Breed's Hill are called **drumlins**, smooth-sloped hills made of glacial sediments. In 1775, the Americans took the high ground and fortified these drumlins by building 6-foot **earthworks**, human-made land modifications, out of the silty glacial soils. This put the Americans in an ideal defensive position overlooking the battlefields.

But taking the high ground doesn't always ensure victory. At Bunker Hill, the Americans held their ground until they ran out of ammunition. As you



Tricky Terrain
Seemingly open stretches of land were obstacle courses for the British. Tall grasses hid rocks and holes. Fences and stone walls also slowed them down.

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have read, the British won the battle, but at a terrible cost. Over 1,000 British soldiers were killed or wounded, compared to over 400 Americans. This showed that the Patriots could hold their own against a better-trained, better-equipped, and far larger British force. Clearly local terrain played a role in how the battle unfolded.

War commanders probably don't think much about the geologic story that shaped their battlefields, but that doesn't mean the geology isn't significant.

THINK LIKE A GEOLOGIST

1. **IDENTIFY MAIN IDEAS AND DETAILS** If you were a commander defending your city, where would you set up troops? Why?
2. **DESCRIBE** What advantages or disadvantages would your local terrain present in battle?
3. **INTERPRET MAPS** Aside from hills, which geologic features do you think affected the battle?

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